

Roll No.

TMA-310

B. TECH. (ECE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022
ADVANCED ENGINEERING MATHEMATICS

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define the harmonic function. Show that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic. Also find the harmonic conjugate function. (CO1)

OR

- (b) Using least square method to fit a curve $y = ax^2 + b$ for the following data : (CO2)

x	y
1	10
3	5
5	20
10	25
15	22

P. T. O.

2. (a) (i) Find the all values of k such that the function $f(z) = e^x(\cos ky + i \sin ky)$ is analytic.

- (ii) Show that the function $f(z) = u + iv$, where :

$$f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$$

is not analytic at $z = 0$.

(CO1)

OR

- (b) Explain the concept of the following terms with the help of *one* example : (CO2)

- (i) Kurtosis
- (ii) Skewness
- (iii) Correlation
- (iv) Regression Analysis

3. (a) Following are the data on the drying time of a certain varnish and the amount of an additive that is intended to reduce the drying time :

Amount of varnish additive groups (x)	Drying time (y in hours)
0	12.0
1	10.5
2	10.0
3	8.0

4	7.0
5	8.0
6	7.5
7	8.5
8	9.5

(i) Fit a second-degree parabola $y = a + bx + cx^2$.

(ii) Use the result of part (i) to predict the drying time of the varnish when 6.5 gm of additive is being used. (CO2)

OR

(b) Using Cauchy's integral formula to compute the value of the integral

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz, \text{ where } C : |z| = 3. \quad (\text{CO1})$$

4. (a) Show that the function $f(z) = u + iv$, where $f(z) = e^{-z^4}$ ($z \neq 0$) and $f(0) = 0$ is not analytic at $z = 0$ although Cauchy-Riemann equations are satisfied at $z = 0$. (CO1)

OR

- (b) The first moments about the working mean 28.5 of a distribution are 0.294, 7.144, 42.409 and 454.98. Calculate the moments about the mean. Also calculate β_1, β_2 and comment upon the Skewness and Kurtosis of the distribution. (CO2)

5. (a) Show that the function $f(z) = z|z|$ is not analytic anywhere. (CO1)

P. T. O.

OR

(b) Evaluate $\oint (5z^4 - 3z^2 + 6) dz$, around the closed curve C, where the curve C is :

(i) the unit circle $|x + iy| = 1$.

(ii) the square PQRS with $P = (0, 0)$, $Q = (1, 0)$, $R = (1, 1)$ and $S = (0, 1)$

(iii) the parabola $C_1 : y = x^2$ and the parabola $C_2 : x = y^2$ joining $(0, 0)$ and $(1, 1)$. (CO1)